Hard submatrices for Boolean 1-partition number

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Editorial note. Faithful transcription from official CCC 2024 CC BY 4.0 PDF; original notation, indexing and dimensions retained. Author TeX was not located. No author proof is replaced or extended; standard cited prior results are retained as prerequisites, and bibliography entries keep their original numbers.

Allowed standard prerequisites (editorial).

- Elementary rank and binomial-coefficient bounds. Author source Sections 3-4

3 A combinatorial lemma (author source pp. 13:3–13:4)

Both Theorems 1 and 2 rely on a simple combinatorial lemma.

Lemma 3. *Let $\mathcal{A} \subseteq 2^{[n]}$ be a family of subsets of $[n]$. Assume that $1 \leq k \leq n$ is such that every k-element subset of $[n]$ is contained in some $A \in \mathcal{A}$. Then there exists a subfamily $\mathcal{A}' \subseteq \mathcal{A}$ of size $|\mathcal{A}'| \leq O(|\mathcal{A}|^{1/k} \log(n/k))$ with $\bigcup \mathcal{A}' = [n]$. In particular, if $|\mathcal{A}| \leq 2^k$ then $|\mathcal{A}'| \leq O(\log n)$.*

Proof. Assume that $|\mathcal{A}| \leq a^k$. Let t be the size of a largest set in $\mathcal{A}$. Then we have

$$\binom{n}{k} \leq a^k \binom{t}{k}.$$

Hence $t \geq \frac{n}{ea}$, using the estimates $(\frac{n}{k})^k \leq \binom{n}{k}$, $\binom{t}{k} \leq (\frac{et}{k})^k$. Take some $A_0 \in \mathcal{A}$ of size t. Let

$$\mathcal{A}_1 := \{A \setminus A_0 : A \in \mathcal{A}\}.$$

Then every subset of $U_1 := [n] \setminus A_0$ of size at most k is contained in some member of $\mathcal{A}_1$. The size of U_1 is at most $n(1 - \frac{1}{ea})$. Similarly, take a largest set A_1 from $\mathcal{A}_1$ and obtain a new family $\mathcal{A}_2 \subseteq 2^{U_2}$ on $U_2 := U_1 \setminus A_1$. After s steps, the size of U_s is at most $n(1 - \frac{1}{ea})^s$ and after $s \leq O(a \log(n/k))$ steps we have $|U_s| \leq k$. This guarantees that the largest set in $\mathcal{A}_s$ is U_s itself and $[n] = \bigcup_{i=0}^{s} A_s$. By construction, every A_i is contained in some element of the original family $\mathcal{A}$. □

Definitions and notation (author source p. 13:3)

Given a Boolean matrix $M \in \{0,1\}^{m\times n}$, let us define its 1-partition number, $\chi_1(M)$, as the smallest s such that M can be written as

$$M = LR, \quad \text{with } L \in \{0,1\}^{m\times s},\ R \in \{0,1\}^{s\times m},$$

where the operations are over $\mathbb{R}$. The definition emphasizes the analogy with rk_+, and χ_1 is also sometimes referred to as binary rank. On the other hand, the phrase "partition number" comes from communication complexity. The name is justified: it is easy to see that $\chi_1(M)$ equals the smallest s such that the 1-entries of M can be partitioned into s 1-monochromatic rectangles (i.e., rank-one Boolean matrices). Finally, when M is viewed as the adjacency matrix of a bipartite graph, $\chi_1(M)$ also appears under the name biclique partition number [3].

Notation. All logarithms are in base 2 and $[n] := \{1,\dots,n\}$.

4 1-Partition number (author source pp. 13:4–13:5)

In this section, we prove Theorem 2. Let M be an $m \times n$ matrix with rows indexed by $[n] = \{1,\dots,n\}$. Given $A \subseteq [n]$, M_A denotes the submatrix obtained by removing the rows outside of A from M. Observe that[1]

$$\chi_1(M_{A_1\cup A_2}) \le \chi_1(M_{A_1}) + \chi_1(M_{A_2}), \tag{1}$$

and so $\chi_1(M_A)$ can be viewed as a subadditive measure on $[n]$ whenever M is fixed. If a matrix M has rank r, its rows are a linear combination of a subset of r rows of M. This means that every column of M is determined by a fixed subset of r coordinates. If M is Boolean, this leads to the following useful fact: if M has distinct columns then $n \le 2^{\mathrm{rk}(M)}$ (similarly for rows).

Lemma 5. *Let M be an $m \times n$ Boolean matrix of rank r. Given $s \in [n]$, let $\mathcal{A}$ be the collection of maximal subsets $A \subseteq [n]$ with $\chi_1(M_A) \le s$ (i.e., $\chi_1(M_A) \le s$ and $\chi_1(M_{A'}) > s$ for every $A' \supsetneq A$). Then $|\mathcal{A}| \le 2^{(r+s)^2}$.*

Proof. Let $v_1,\dots,v_n \in \mathbb{R}^m$ be the columns of M. Given $L \in \{0,1\}^{m\times s}$, let

$$L^* := \{i \in [n] : \exists y \in \{0,1\}^s\ v_i = Ly\}.$$

Let $\mathcal{L} := \{L^* : L \in \{0,1\}^{m\times s}\}$. We claim that $\mathcal{A} \subseteq \mathcal{L}$. If $\chi_1(M_A) \le s$, we can write $M_A = LR$ with $L \in \{0,1\}^{m\times s}$ and $R \in \{0,1\}^{s\times|A|}$. This means that every v_i, $i \in A$, is a Boolean linear combination of the columns of L and $A \subseteq L^*$. Furthermore, if A is maximal, we must have $A = L^*$. We now want to estimate the size of $\mathcal{L}$. The set L^* consists of indices $i \in [n]$ so that there exists $x \in \mathbb{R}^n, y \in \mathbb{R}^s$ satisfying

$$Mx - Ly = 0 \tag{2}$$

[1] If A_1, A_2 are disjoint, this is quite obvious. Otherwise consider $A_1, A_2 \setminus A_1$.

such that $y \in \{0,1\}^s$ and x is the i-th unit vector. Since M has rank r and L has rank at most s, the system (2) is equivalent to a subsystem of $t := \min((s+r), m)$ equations. These correspond to rows of the matrix (M, L). Hence, in order to determine L^*, it is sufficient to specify a t-element subset of $[m]$ together with the $t \times s$ submatrix of L. This gives the estimate

$$|\mathcal{L}| \leq \binom{m}{t} 2^{ts} \leq 2^{t(s+\log m)}.$$

Finally, we can assume that M has distinct rows and so $\log m \leq r$, obtaining the bound $2^{(r+s)^2}$. □

Theorem 2 (restated; author source p. 13:5). *Let M be an $m \times n$ Boolean matrix with $n \geq 2$. Then for every $k \leq n$, M contains an $m \times k$ submatrix of k columns with 1-partition number $\Omega(\min(\sqrt{k}, \frac{\chi_1(M)}{\log n}))$.*

Proof. Let r be the rank of M. We will assume $r \leq \frac{k^{1/2}}{2}$. Otherwise, observe that M contains a full rank $r \times r$ submatrix, χ_1 is lower-bounded by rank, and the conclusion of the theorem follows. Let s be the maximum $\chi_1(M_A)$ over all $A \subseteq [n]$ of size k. Let $\mathcal{A}$ be the family from the previous lemma. If $|\mathcal{A}| \geq 2^k$, we have $2^k \leq 2^{(s+r)^2}$ and therefore $s \geq \frac{k^{1/2}}{2}$ from the assumption on r. Assume $|\mathcal{A}| \leq 2^k$. By Lemma 3, there exists a subfamily $\mathcal{A}' \subseteq \mathcal{A}$ of size $O(\log n)$ which covers $[n]$. Using (1), this implies $\chi_1(M) \leq O(s \log n)$ and so $s \geq \Omega(\chi_1(M)/\log n)$. □

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